

Vector Floors: Farkas-Certified Obstruction under Multiple Sustainability Constraints

Edition 2, revised after a joint audit

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Abstract

Sustainability constraints are rarely single floors: minimum stocks, maximum emissions, service levels, and institutional limits bind simultaneously. The calculus states its certificates for general vector constraint sets $\mathcal{V} = \{x : q_j(x) \geq 0, j = 1, \dots, m\}$ (Abaee 2026a, Section 2.2), but its worked certificates and audited instances are carried by a single scalar margin (Section 3.4’s two-floor certificate there). This paper, the D4 item of the programme’s generalization catalogue, works the vector layer as such: the stacked safe-action system is infeasible precisely when a Farkas multiplier vector exists, so the vector obstruction itself is a checkable object of rational arithmetic; the new content over the source paper is the dual-certificate reading, the floor-monotonicity count on the audited pair family, and the three-floor support instance. On the programme’s two-patch working system (Abaee 2026a, Section 8) the dynamic vector-floor reading reproduces the audited structure verdicts (blind and aggregate readings nonviable; z_1 -only, z_2 -only, and full readings viable), and adding a floor strictly shrinks the viable pair family — from 36 of 36 viable pairs on one floor to 28 on two, the pairs lost being exactly those containing the dynamically dead corner (1, 1). A three-floor stacked instance exhibits the support reading of the multipliers (the certificate’s carrier is a minimal infeasible subsystem: inactive row, multiplier zero). All claims are machine-verified in exact rational arithmetic; the verification script ships with the paper.

Introduction and relation to the programme

The programme’s completed items — the obstruction-calculus manuscript with its four certificate families, the continuous-to-finite companion bridge, the viable-selector unification, the stochastic selector, and the five-layer successor absorbing general information structures and exit-time values — define vector constraint sets from the outset ($\mathcal{V} = \{x : q_j(x) \geq 0, j = 1, \dots, m\}$, Abaee 2026a, Section 2.2), and the polyhedral Farkas equivalence for the stacked safe-action system is already stated there (Section 3.2),

with a worked two-floor certificate (Section 3.4). What the completed items do not do is carry the vector layer as an object of computation: the audited instances are scalar-margin instances, and the catalogue’s D4 proposal — real constraint systems are vectors of floors — is left unexecuted. The present paper executes exactly that: the vector common-action correspondence, the stacked certificate as a checkable dual object, the floor-monotonicity count on the audited pair family, and the three-floor support instance. Nothing in the completed items is modified; the additions are computational, and each is positioned against its source passage.

The vector-floor model

Let the state space carry m floors $q_1, \dots, q_m$ with safe set

$$\mathcal{V} = \{x : q_j(x) \geq 0 \text{ for every } j\}.$$

For a state x with instrument set $U(x)$ and discrete-time transition $x^+ = F(x, u, d)$, the *vector safe-action set* collects the instruments that keep every floor non-negative under every admissible disturbance:

$$\mathcal{R}_{\mathcal{V}}(x) = \{u \in U(x) : x^+ \in \mathcal{V} \text{ for all successors } x^+\}.$$

For a belief B (the set of states compatible with the reading), the common-action condition with per-cell semantics reads $\mathcal{R}_{\mathcal{V}}^B(B) = \bigcap_{x \in B} \mathcal{R}_{\mathcal{V}}(x)$ within each indistinguishability cell, exactly as in the successor’s read-then-act convention. The vector obstruction is the static statement

$$\mathcal{R}_{\mathcal{V}}^B(B) = \emptyset \implies B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V}),$$

where $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$ is the observation-based viability kernel under structure $\mathcal{I}$ in the programme’s notation (Abaee 2026a, Section 2.4): a single moment without a jointly floor-preserving instrument certifies nonviability, no matter how capable the future.

Proposition 1 (Branchwise/joint gap for vector floors). (The vector instance of the common-action obstruction, Abaee 2026a, Theorem 2; the vector-specific content is the stacked dual layer of Proposition 2 and the floor-monotonicity of E4.) If for each state $x \in B$ separately $\mathcal{R}_{\mathcal{V}}(x) \neq \emptyset$ but $\bigcap_{x \in B} \mathcal{R}_{\mathcal{V}}(x) = \emptyset$, then every branch of B is individually floor-servable while the belief is vector-obstructed: the information constraint, not any single floor, is the cause of nonviability.

Proof. The branchwise hypothesis provides, for each x , an instrument u_x keeping x in $\mathcal{V}$ under *all* floors jointly; each branch is thus servable in isolation. The joint hypothesis denies the existence of one instrument admissible for all x simultaneously, which is precisely the per-cell read-then-act requirement; any observation-based policy must move the whole cell with one instrument and therefore fails at the first step. The cause cannot be attributed to any single floor’s infeasibility (every branch is jointly floor-servable), nor to the future (the failure occurs in one step): it is the pairing of one instrument with several states, i.e. the information constraint. $\square$

The Farkas certificate

Let the floors and the instrument set be polyhedral, so that each safe-action set is $\mathcal{R}_V(x_i) = \{u : A_i u \leq b_i\}$ with rational data. The common safe-action system stacks to $Au \leq b$, and:

Proposition 2 (Obstruction as dual infeasibility). The stacked system $Au \leq b$ is infeasible if and only if there exists a rational $\lambda \geq 0$ with $\lambda^\top A = 0$ and $\lambda^\top b < 0$ (Farkas’ lemma). Consequently the vector obstruction admits a certificate that is itself checkable in exact rational arithmetic: given λ , the three identities are verifiable by Fraction arithmetic with no solver and no search.

Proof. Farkas’ lemma gives the equivalence. Checkability: the certificate side requires evaluating two matrix products and one inequality — exact operations on rationals. The infeasibility side is certified, not searched: $\lambda^\top A = 0$ and $\lambda^\top b < 0$ with $\lambda \geq 0$ yield $0 = \lambda^\top (Au) \leq \lambda^\top b < 0$ for every candidate u , a contradiction. $\square$

Remark 1 (Support reading). In the stacked system of a three-floor instance below, the certified multiplier assigns weight 0 to a redundant row: the certificate’s *support* is a minimal infeasible subsystem of the stacked rows (an infeasible system admits no primal solution, so the reading is one of dual support, not of primal activity). The obstruction is carried by exactly the floors that bind.

Verified instances

All of the following are re-derived and checked by the verification script (8/8 exact check families; standard library, Fraction arithmetic, deterministic).

E1 (stacked equivalence, both directions). The two-floor instrument instance $\mathcal{R}_1 = \{u : u \leq \frac{2}{5}\}$, $\mathcal{R}_2 = \{u : u \geq \frac{3}{5}\}$ stacks to rows $u \leq \frac{2}{5}$, $-u \leq -\frac{3}{5}$; the multiplier $\lambda = (1, 1)$ satisfies $\lambda^\top A = 0$ and $\lambda^\top b = \frac{2}{5} - \frac{3}{5} = -\frac{1}{5} < 0$, certifying infeasibility; a 101-point rational grid confirms no u satisfies both. The feasible contrast $\{u \leq \frac{2}{5}\} \cap \{u \geq \frac{1}{5}\}$ (with a redundant third row $u \leq \frac{4}{5}$) admits no certificate — the exact case analysis forces $\lambda_1 = \lambda_2$ and then $\lambda^\top b \geq 0$ — and the witness $u = \frac{1}{5}$ is exhibited. Both directions of Proposition 2 are thereby exercised; the instance is the unnormalized form of the worked certificate of Abaee 2026a, Section 3.4 (there with the multiplier normalized to $\|\lambda\|_1 = 1$), and the feasible contrast is the direction that source states but does not exercise.

E2 (the belief-level gap). On the belief merging a state with floor-set $\{u \leq \frac{2}{5}\}$ with a state with floor-set $\{u \geq \frac{3}{5}\}$: branchwise witnesses $u = \frac{2}{10}$ and $u = \frac{6}{10}$ exist; the joint set is empty with the E1 certificate. Proposition 1 fires exactly.

E3 (dynamic vector floors). On the programme’s two-patch system — patches $i \in \{1, 2\}$ with stocks $z_i \in \{0, 1, 2, 3\}$, regeneration +1 per step capped at 3, damage 2 on the unprotected patch, instruments $u \in \{1, 2\}$ protecting one patch; defined in Abaee 2026a, Section 8, and restated with full transition data in the verification script — the safe set is already a vector object $q_1 : z_1 \geq 1$, $q_2 : z_2 \geq 1$ ($m = 2$) served by one shared instrument. Per-cell read-then-act viability over the audited beliefs $\{(1, 2), (2, 1)\}$ and $\{(1, 2), (2, 1), (2, 2)\}$ reproduces the audited verdicts under five information structures: blind nonviable, aggregate nonviable, z_1 -only viable, z_2 -only viable, full viable — the successor’s structure table re-read as a statement about two floors and one instrument.

E4 (floor monotonicity). The *viable pair family*: the $\binom{9}{2} = 36$ two-state beliefs of the nine floor-safe states, a pair being viable when both of its states are (full observation splits the pair into its branches). Adding the second floor strictly shrinks the family from all 36 to 28: under two floors the corner $(1, 1)$ has *no* instrument keeping both floors (one shared instrument, damage 2 on the other patch), so $(1, 1)$ is dynamically nonviable and exactly the 8 pairs containing it are lost. Dropping the second floor (q_1 only) re-admits all 36, setwise, with strict separation witnessed by $\{(1, 1), (1, 2)\}$: viable on one floor, nonviable on two. Adding floors destroys viability — the marginal cost of each additional institutional constraint is a computable set.

E5 (exact multiplier recheck). Every multiplier used above is re-validated componentwise ($\lambda \geq 0$; $\lambda^\top A = 0$; $\lambda^\top b < 0$) in Fraction arithmetic.

E6 (three floors, complementarity). The stacked system $\{u \leq \frac{2}{5}\} \wedge \{u \geq \frac{3}{5}\} \wedge \{u \leq \frac{9}{10}\}$ is infeasible with certificate $\lambda = (1, 1, 0)$: the inactive third row carries multiplier zero, exhibiting the support reading of Remark 1 (the script further certifies minimality: every two-row subsystem containing row 2 or row 3 is feasible).

What is not claimed

The dynamic vector-floor theory is not developed beyond the per-cell re-reading of the audited systems: no vector-valued comparison functions, no multi-floor dynamic programming, no Pareto readings of trade-offs between floors. The Farkas layer is static; the companion bridge’s continuous-to-finite transfer and the successor’s timing layer are cited, not re-derived. Non-convex instrument codices invalidate Proposition 2 (Farkas’ lemma is a convexity statement); the monitoring-design companion of this catalogue records the exact non-convex counterinstance. Sustainability floors with uncertain thresholds are treated in the programme’s buffered-viability companion, not here. The dynamic vector layer is likewise recorded, not developed: the first-breach timing bound for floors with margins ε_j — exit by $\min_j \inf_V q_j / \varepsilon_j$ — and the multi-face Isaacs reading at corners where several floors are simultaneously active belong to the successor’s timing layer; so does the distinction between cross-floor infeasibility at one state and cross-state infeasibility under merged readings, which Proposition 1 isolates in its static, one-step form.

Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The verification script

`vector_floor_certificates_v2_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all eight check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review of the first edition; the joint audit of the first edition and this revision were prepared with further AI assistance, under the programme’s verification discipline.